MILITARY SPECIFICATION
CONTAINER, SHIPPING AND STORAGE
GUIDED MISSILE CNU-159/E

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of one type of reusable shipping and storage container for the PHOENIX Missile AIM-54A Propulsion Section.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS:

FEDERAL

FED-STD-101	Preservation, Packaging & Packing Materials: Test Procedures

MILITARY

MIL-STD-105	Sampling Procedure & Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment & Storage
MIL-STD-648	Design Criteria for Specialized Shipping Containers

SPECIFICATIONS:

MILITARY

MIL-D-3464	Desiccants, Activated, Bagged

PUBLICATIONS:

DL 67A246F1	Container, CNU-159/E

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Air Systems Command, Engineering Division, Standardization Section, Code: AIR-52021, Department of the Navy, Washington, D.C. 20361, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 8140

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 General Requirements

3.1.1 Construction. The CNU-159/E Container shall be manufactured in accordance with NAVAIR Data Lists (DL) 67A246F1 and all documents referenced thereon.

3.1.2 Conflicting Requirements. Conflicting requirements arising between this specification or any specification, publication or drawing listed herein, shall be referred in writing to the procuring agency or Contracting Office for interpretation and clarification.

3.1.3 Preproduction Sample. Unless otherwise specified in the contract or purchase order (see 6.2), two (2) preproduction samples of the container shall be manufactured using the methods and procedures proposed for production. The sample will be tested as specified in Section 4 herein for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of this specification.

3.2 Performance Requirements and Product Characteristics. Containers covered by this specification shall be capable of meeting the following performance requirements and product characteristics.

3.2.1 Leakage. The pressure drop shall not exceed .05 psig pressure drop in one hour when subjected to the leak test described in 4.4.0.

3.2.2 Shock and Vibration. There shall be no evidence of structural failure or inability of the container to protect its contents when the container is subjected to the following tests:

a. Low Order Repetitive Shock - 4.4.1

b. Rotational Drop - 4.4.2

c. Impact - 4.4.3

d. Flat Drop Test - 4.4.4

e. Vibration Fatigue - 4.4.5

3.2.2 Stacking. There shall be no evidence of breakage or permanent distortion, temporary distortion damaging the contents, or development of an unstable configuration when the container is subjected to the stacking test described in 4.4.6.

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3.2.4 Hoisting. There shall be no evidence of permanent deformation of the fittings, handles, or the structure adjacent thereto when the container is subjected to the hoisting tests described in 4.4.7.

3.3 Workmanship. Workmanship in the fabrication and assembly of containers, shall be of sufficiently high quality to assure that completed equipment will comply with all requirements of this specification. Particular attention shall be paid to plating, bonding, painting, welding, riveting, marking and to producing parts free of burrs and sharp edges. The assembled containers shall be thoroughly clean of all dirt, chips, and other foreign matter. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspections. Inspection of the container shall be classified as follows:

a. First Article Inspection (see 4.4)

b. Quality Conformance Inspection (see 4.5)

4.3 First Article

4.3.1 First Article Sample. Two (2) preproduction samples of containers manufactured in accordance with 3.1.1 and 3.1.3 shall be subjected to the preproduction tests detailed in 4.4. Acceptance of the preproduction samples shall be based on no defects in the samples.

4.4 First Article Inspection. Preproduction samples, after satisfactorily passing the examinations of 4.5.1 shall be subjected by the Naval Weapons Station Earle, Colts Neck, New Jersey, 07722, to the following tests:

4.4.0 Leakage Test. The loaded container shall be leak tested in accordance with the pneumatic pressure technique of FED-STD-101 Method

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5009. The container shall be pressurized to 2.0 ± 0.5 psig and shall meet the requirements of 3.2.1.

4.4.1 Low Order Repetitive Shock Test. The loaded container shall be tested at ambient temperature in accordance with MIL-STD-648 paragraph 5.3.2 and shall meet the requirements of 3.2.2.

4.4.2 Cornerwise (Rotational) Drop Test. The loaded container shall be tested at -20°F and +140°F in accordance with FED-STD-101 Method 5005 level B and shall meet the requirements of 3.2.2.

4.4.3 Incline Impact Test. The loaded container shall be tested at -20°F and +140°F in accordance with FED-STD-101 Method 5023 with optional timber except endwise impacts shall be at a velocity of 10 feet per second and shall meet the requirements of 3.2.2.

4.4.4 Flat Drop Test. The loaded containers shall be subjected to an 18-inch free fall flat drop at ambient temperature and shall meet the requirements of 3.2.2.

4.4.5 Vibration Fatigue Test. The loaded container shall be tested at ambient temperature in accordance with MIL-STD-648 paragraph 5.3.2 and shall meet the requirements of 3.2.2.

4.4.6 Stacking Test. The loaded container shall have a superimposed load of 11,500 pounds applied for a period of 1 hour on its load bearing areas and shall meet the requirements of 3.2.3.

4.4.7 Hoisting Tests. To verify the strength of the hoisting fittings, the loaded container shall be tested in accordance with MIL-STD-648 paragraph 5.8.1 and shall meet the requirements of 3.2.4.

4.4.8 Leakage Test. The leakage test of 4.4.0 shall be repeated and shall meet the requirements of 3.2.1.

4.5 Quality Conformance Inspection

4.5.1 Examination. Containers shall be examined for compliance with the requirements of NAVAIR Data List DL 67A246F1 (see 4.5.2).

4.5.2 Each container shall be inspected utilizing a GFE gage, P/N NWHC 7796 [provided by WPNSTA Earle (NWHC)] to verify the contour of missile section supports item 4, 30003-67A246F24 and straps 30003-67A246D16. Contour of support and strap in the closed (latched) configuration shall be such that no clearance greater than 0.060" ± .005" shall exist between supports/straps and gage.

4.5.3 Lot and Acceptable Quality Level. All containers of the same design and configuration, manufactured under the same conditions and presented for acceptance at one time, shall be considered an inspection lot for quality conformance tests. Inspection shall be conducted in accordance with MIL-STD-105 at Level II with an Acceptable Quality Level (AQL) of 1.0.

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4.5.4 Test and Equipment and Test Facilities. Unless otherwise specified the manufacturer (see 4.1) shall furnish and maintain all normal test equipment, facilities and personnel necessary for performing all quality conformance tests. The test equipment shall be adequate in quantity, and unless definite requirements are specified, tney shall be of sufficient accuracy and quality to permit performance of the required quality conformance test.

4.5.5 Test Conditions. Unless otherwise specified in a particular test, all tests shall be performed at a temperature between 50°F and 90°F with ambient humidity.

4.5.6 Tests. Each container manufactured shall be subjected to the pressure (leak) test of 4.4.0 except containers shall show zero leakage for 5 minutes.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Twenty-four (24) units of fresh Desiccant, MIL-D-3464, shall be placed in each container and the container cover secured.

5.2 Packing. The container requires no packing for delivery. However, all loose interior pieces and all latches shall be tightly secured.

5.3 Marking. In addition to any special marking required by the contract or purchase order, containers shall be marked in accordance with the applicable marking drawing and MIL-STD-129.

6. NOTES

6.1 Intended Use. Containers supplies under this this specification are intended for use in the shipment and storage of PHOENIX Missile AIM-54A Propulsion Sections.

6.2 Ordering Data. Procurement documents shall specify:

a. Title, number and date of this specification.

b. First Article sample size, if different (see. 4.3.1).

c. Activity designated to perform testing (see 4.4).

Preparing activity:
Navy-AS

Project 8140-N422

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-C-85123(AS)
CONTAINER SHIPPING AND STORAGE, GUIDED MISSILE CNU-159/E

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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